Unit IV

Fuels

Definition and classification of fuels, high & low Calorific value.

Liquid fuels: Petroleum and its refining, distillation of petroleum (Kerosene oil, Diesel and Petrol), Power alcohol, Knocking, Anti-knocking agents, Octane number and Cetane number.

Gaseous fuels: Coal gas, Oil gas, Water gas, Producer gas, Bio gas, LPG and CNG. Numerical Problems based on topics.

Introduction

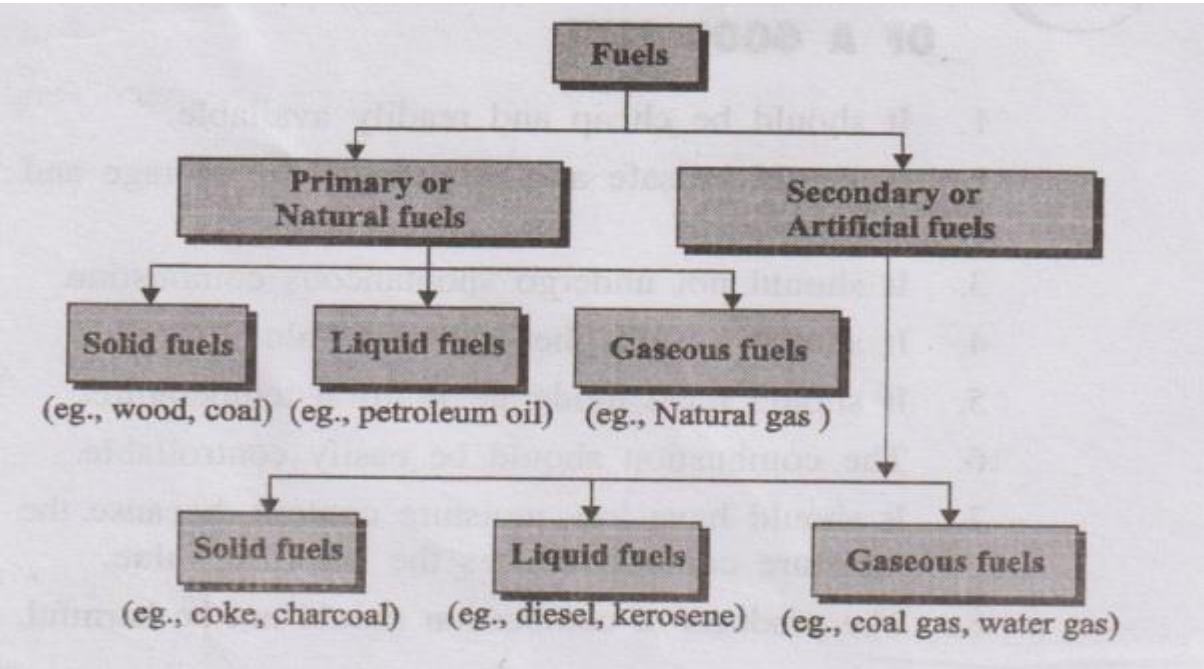
A fuel is a substance that contains carbon and hydrogen undergoes combustion in presence of oxygen to gives large amount of energy.

$$\text{Fuel} + O_2 \rightarrow CO_2 + H_2O + \text{Energy}$$

Classification of Fuel

On the basis of occurrence fuel is classified into two categories; natural or primary fuels and artificial or secondary fuels.

- Natural/primary fuels: These fuels are naturally present.
- Artificial/ secondary fuels: They are synthesized by primary fuels.



Characteristics of good fuel

- Fuel should have high calorific value.
- Must have moderate ignition temperature.
- Fuel should have low moisture content.
- Available in bulk at low cost, should not burn spontaneously.
- Fuel should burn efficiently, without releasing hazardous pollutants.
- Handling, storage and transportation should be easy.

Unit of heat

- **Calorie:** It is the amount of heat required to raise the temperature of 1 gram of water through one degree centigrade.
- **British Themal Unit (BTU):** it is the amount of heat required to raise the temperature of 1 pond of water to one degree Fahrenheit.
  1 B.T.U. = 252 cal = 0.252 kcal 1 kcal = 3.968 B.T.U.
- **Centigrade Heat Unit (CHU):** it is define as the amount of heat required to raise the temperature of 1 pond of water to one degree centigrade.
  1 kcal = 3.968 B.T.U. = 2.2 C.H.U.

Calorific value

Calorific value of fuel can be define as the amount of heat evolved when one unit mass or volume of the fuel undergoes completely combustion in presence of oxygen.

- **High or gross calorific value (HCV or GCV):** it is defined as amount of heat evolve when one unit mass or volume of the fuel is completely burnt and combustible products are cooled to room temperature (25°C or 77°F).
- **Low or net calorific value (LCV or NCV):** it is defined as amount of heat evolve when one unit mass or volume of the fuel is completely burnt and combustible products are permitted to escape. Therefore net calorific value is lower than gross calorific value.
  LCV = HCV – latent heat of water vapour
  LCV = HCV – mass of hydrogen × 9 × latent heat of steam (587 kcal/kg)

One part by mass of hydrogen produced nine parts by mass of water molecule. Therefore,

LCV = HCV – H/100 × **9** × 587 kcal/kg
H = percentage of hydrogen in fuel

Petroleum composition

Depending upon the number of carbon and hydrogen atoms the petroleum products are classified into different groups. They are:

(i) Paraffin series (C_nH_{2n+2}) (ii) Olefin series (C_nH_{2n}) (iii) Naphthene series (C_nH_{2n}) (iv) Aromatic series (C_nH_{2n-6})

Family of Hydrocarbons	Chemical Formula	Molecular Structure	Saturated / Unsaturated	Stability
Alkanes (paraffins)	C_nH_{2n+2}	Chain	Saturated	Stable
Olefin	C_nH_{2n}	Chain	Unsaturated	Unstable
Naphthenes	C_nH_{2n}	Ring	Saturated	Stable
Aromatics	C_nH_{2n-6}	Ring	Highly Unsaturated	More Unstable

Classification of petroleum

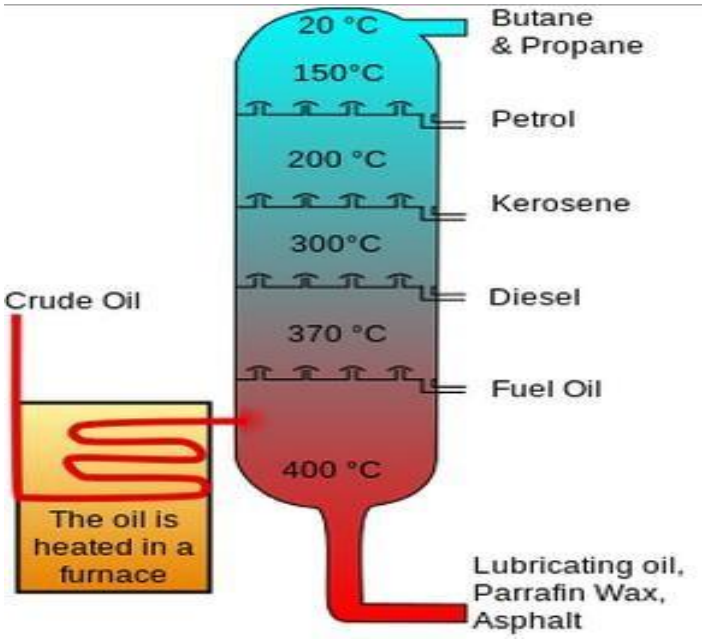
For several decades now the crude oil or petroleum industry has classified the raw crude geographic region. Further classification of petroleum, derived from the density of the raw petroleum (API gravity) and its various non-hydrocarbon components (especially sulphur), is then added to the geographic designation.

In general, if the crude oil contains high levels of sulphur the petroleum classification is termed ‘sour’ if it has relatively low levels of sulphur the petroleum classification is termed ‘sweet'. If the raw petroleum is of a high density then the petroleum classification is termed ‘heavy' and if it is of a low density the petroleum classification is termed 'light'. Density of oil is determined by the length of the hydrocarbons it contains. If it contains a great deal of long-chain hydrocarbons, the petroleum will be denser. If it contains a greater proportion of short-chain hydrocarbons it will be less dense.

The API gravity is used to classify oils as light, medium, heavy, or extra heavy. As the “weight” of oil is the largest determinant of its market value, API gravity is exceptionally important. The API values for each “weight” are as follows:

Classification	API Value
Light	API > 31.1
Medium	22.3<API < 31.1
Heavy	API < 22.3
Extra Heavy	API < 10.0

API Gravity: API stands for the American Petroleum Institute, which is the major United States trade association for the oil and natural gas industry. One of the most important standards that the API has set is the method used for measuring the density of petroleum. This standard is called the API gravity.



Important fraction of Petroleum and their uses

- **Petroleum gas:** Liquefied Petroleum Gas or LPG is a flammable liquid that is a combination of various hydrocarbons. The most common forms of LPG are propane and butane. It is also produced by refining crude oil. Its boiling range is less than 25 ^{0}C. It is used as a fuel in homes and industry.
- **Petrol:** Petrol is also referred to as Gasoline and is a transparent liquid obtained from the fractional distillation of crude oil. It consists of organic compounds. It is used in vehicles, electrical generators, compressors etc. It is also used as a solvent for dry cleaning.

  It contains a mixture of hydrocarbons between pentane to octane (C_5H_{12} to C_8H_{18}).

  It is volatile and inflammable.

  Average Composition C = 84%, H=15%, O+S+N =1%

  Calorific Value = 11,250 Kcal / Kg.

  Uses:

  It is used as a fuel for internal combustion engines of automobiles.

  It is used as a dry-cleaning agent.
- **Kerosene:** Kerosene, also called paraffin, is a combustible liquid containing hydrocarbons. Kerosene is used to power jet engines, for cooking, heating, lighting fuels and toys. Other applications of Kerosene are as a pesticide, as a solvent, lubricant, x-ray crystallography and so on.

It consists of hydrocarbons between decane to hexadecane ($C_{10}H_{22}$ to $C_{16}H_{34}$).

Average Composition, C = 84 % , H = 16%, S < 0.1%

Calorific value = 11,100 Kcal/ Kg

Uses:

It is used as a domestic fuel in stoves.

It is used as jet engine fuel for making oil gas.

- **Diesel:** Diesel is obtained by fractional distillation of crude oil. . The quality of diesel fuel is measured in terms of Cetane numbers. It is used as a fuel in heavy motor vehicles. It is also used to run pump sets for irrigation in agriculture and in electric generators.

  It contains a mixture of hydrocarbons between pentadecane to octadecane ($C_{15}H_{32}$ to $C_{18}H_{38}$).

  Average composition: C = 85%, H = 12%, Rest = 3%

  Calorific Value = 11,000 Kcal/kg

Uses:

It is used as a fuel in diesel engine.

- **Lubricating oil:** A good lubricant has a high boiling point and viscosity. It is used for lubrication in machines and engines. Lubricants are also used as transmit power, transfer heat, prevent corrosion and rusting etc.
- **Paraffin Wax:** Paraffin is a waxy solid that is used as a lubricant. It is typically a solid at room temperature. It is used for making candles, vaseline, ointments, wax paper, and grease.
- **Bitumen:** It is used for road surfacing. It is also used for water-proofing the roofs of the building. It is used in making black paints.

Power Alcohol

Power alcohol is a type of fuel. So, it contains materials that can easily undergo combustion reaction. Power alcohol is used in some types of automobiles as a fuel.

Power alcohol is the mixture of absolute alcohol and petrol. In fuel engines, various types of fuels are used. Alcohol is also added with petrol in some amounts in order to get some benefits. When Absolute alcohol and Petrol (20 : 80) mixture is used, the mixture is called

power alcohol. Power alcohol has a good anti-knocking property and its octane number is 90 which is very good.

Why absolute alcohol is added to petrol to make power alcohol?

- Ethyl alcohol can easily be produced by fermentation processes and other many such processes. It is obtained as a side product in many reactions as well. And on the other hand, petrol is obtained from crude oils and as we know that is a limited source of energy, prices of them are also very high. Crude oils are not found everywhere. That's why the addition of absolute alcohol will make fuel cheaper.
- As power alcohol also contains oxygen atoms, it helps the engine to make complete combustion of petrol and reduces the production of harmful CO gas.

KNOCKING

In an internal combustion engine, a mixture of gasoline vapour and air is used as the fuel. After the ignition by spark in the cylinder, the flame should spread rapidly and smoothly through the gaseous mixture. The gases evolved during ignition force piston down due to high pressure, thus providing the power stroke.

The ratio of the gaseous volume in the cylinder at one end of the suction stroke to that at the other end of the compression stroke of the piston is known as compression ratio.

Higher the compression ratio higher the efficiency of engine. If the compression ratio exceeds a certain limit, full air mixture may be heated to a temperature greater than its ignition temperature and spontaneous combustion occurs even before sparking. This is known as pre ignition.

Sometimes, due to the presence of some constituents in the gasoline used, the rate of oxidation becomes great that the portion of the fuel air mixture gets ignited instantaneously producing an explosive violence, known as knocking. This results in loss of efficiency. The knocking is related to the structures of the hydrocarbons in petrol.

The structures and knocking are in the following order:

Straight chain paraffins > branched chain paraffins olefin > cyclo-paraffins > aromatics Thus, olefins of the same carbon chain length possess better anti-knock properties than the corresponding paraffins.

Some of the effects of knocking or detonation are:

- Carbon deposits on liners and combustion chamber
- Mechanical damage
- increase in heat transfer

- Noise and roughness
- decrease in power output and efficiency
- preignition

The knocking can be controlled or even stopped by the following methods:

- increasing engine r.p.m
- reducing pressure in the inlet manifold by throttling
- Retarding spark
- Making the ratio too lean or rich, preferably latter.
- Water injection increases the delay period as well as reduces the flame temperature.

Use of high octane fuel can eliminates detonation. High octane fuels are obtained by adding additives known as dopes like tetraethyl lead, benzol, xylene to petrol

Antiknocking Agent:

The anti-knocking agents are the compounds added to the petrol engine in limited quantities to reduce the propensity to knocking. This compound is intended to reduce emissions of carbon monoxide and ozone in the environment. $(C_2H_5)_4Pb$, Tetraethyl lead (TEL) is octane rating booster that allowed a significant increase in engine compression. This in turn caused the vehicle efficiency and fuel economy to improve.

Octane number:

Octane number is measure of the ability of a fuel to resist knocking when ignited in a mixture with air in the cylinder of an internal-combustion engine. The knocking characteristic of a fuel can be easily expressed by octane number. The anti- knocking value of n-heptane is taken as 0 (zero) because n-heptane knocks very badly. Whereas the anti-knock value of iso- octane is approximately taken as 100 because iso-octane knocks very little. Actually the octane number is the percentage of iso-octane in a mixture of n-heptane in order to matches the knocking characteristics of the fuel. In this way, an “80-octane” fuel is one which has the same combustion characteristics as an 80 **:** 20 mixture in iso-octane and n-heptanes. Gasoline with octane rating as high as 135 are used for aviation purposes. The octane number of poor fuels can be raised by the addition of extremely poisonous materials as tetra ethylene lead $(C_2H_5)_4Pb$ and diethyl-telluride $(C_2H_5)_2Te$.

Isooctane = octane number 100 (good fuel)

Octane = octane number zero (bad fuel)

Lead petrol:

The variety of petrol in which tetra ethyl lead is added, it is leaded petrol.

Octane rating:

It has been found that n-heptane, Knocks very badly and hence, its anti-knock value has arbitrarily been given zero. On the other hand, isooctane (2, 2, 4 – trimethyl pentane). It gives very little knocking, so its anti-knock value has been given as '100'. Thus, octane number (or rating) of a gasoline (or any other internal combustion engine fuel) is the percentage of isooctane in a mixture of isooctane and n-heptane, which matches the fuel under test in knocking characteristics. In this way, an "80-octane" fuel is one which has the same combustion characteristics as an 80:20 mixture of isooctane and n-heptane.

Advantages:

Usually petrol with low octane number is not good quality petrol. It often knocks (i.e., produces huge noise due to improper combustion). As a result of knocking, petrol is wasted; the energy produced cannot be used in a proper way.

When tetra ethyl lead is added, it prevents knocking, there by saves money and energy. Usually 1 to 1.5 ml of TEL is added per 1lit of petrol.

The mechanism of action is as follows:

First TEL will be transformed into finely divided particles of PbO which looks like a cloud. This takes place in the cylinder. Then the PbO particles react with hydrocarbon peroxide molecules formed, thus slowing down the oxidation process and prevent early detonation. Thus either knocking may be stopped or greatly reduced.

Disadvantages:

Deposits of PbO are harmful to engine. So PbO must be eliminated from the engine. For this purpose, little amount of ethylene dibromide is added to petrol. It converts the harmful PbO to volatile $PbBr_2$ and eliminated through exhaust. Presence of any sulphur compounds reduces the efficiency of TEL.

Cetane number:

The Cetane Number (CN) is a measure of a fuel's ignition delay, the time period between the start of injection and the first identifiable pressure increase during combustion of the fuel. In a particular diesel engine, higher cetane fuels will have shorter ignition delay periods than lower cetane fuels. The cetane number is calculated by determining rich mixture of cetane (hexadecane) (CN = 100) and iso-cetane (2, 2, 4, 4, 6, 8, 8-heptamethylnonane)(CN = 0)

The main difference between the cetane and octane number of a fuel is that the octane rating represents how well a fuel can resist pre-ignition due to compression which ensures

that the fuel ignites only from a spark from the spark plug. However, the cetane number measures the delay in the ignition time of the fuel.

Composition and uses of some gaseous fuels:

- **Water Gas**

- It is a mixture of combustible gases CO and H_2 with a little quantity of non-combustible gases like CO_2 and N_2.
- The average composition of water gas is H_2= 51 %, CO = 41 %, CO_2 = 4% & N_2 = 4%,
- Its calorific value is 2800 Kcal / m^3 .

Uses: It is used:

- as a source of H_2 Gas.
- as a fuel.
- as an illuminating gas.
- for welding purposes.

- **Producer Gas:**

- It is a mixture of combustible gases, CO and H_2 with large quantities of non-combustible gases CO_2 and N_2.
- The avg. composition of producer gas is CO = 22-30%, H_2 = 8-12 %, N_2 = 52-55 % & CO_2 = 3%
- Its calorific value is 1,300 Kcal /m^3.

Uses: It is used:

- For heating open-hearth furnaces in steel & glass manufacture, muffle furnace in coke & coal gas manufacture.
- As a reducing agent in metallurgical operations.

- **Coal gas:**

- It is a mixture of a number of hydrocarbons along with N_2, H_2, CO & CO_2.
- It is a colourless gas and burns with a sooty flame.
- The average composition of coal gas is H_2 = 40%, CH_4 = 32%, CO = 7%, C_2H_2 = 2%, C_2H_4 = 3%, N_2 = 4%, CO_2 = 1% & rest = 11%
- Its calorific value is 4900 Kcal /m3.

Uses: It is used:

- as a fuel.
- as a reducing agent in metallurgical operations.

- as an illuminate.

➢ **LPG (Liquified petroleum gas):**

- It is mainly C3, C4 hydrocarbons of alkane & alkene.
- It is highly inflammable.
- It is colourless and odourless but a smelling agent called ethyl mercaptan (C_2H_5SH) is added to it to detect the leakage.
- The average composition of LPG is n-butane = 27%, iso-butane = 25%, butane = 43%, propene = 2.5% & propane = 2.5%.
- Its calorific value is 27,800 Kcal /m^3.

Uses: It is used:

- As a domestic fuel.
- As an industrial fuel.
- As a vehicular fuel.

➢ **CNG (Compressed Natural Gas):**

- It is a colourless, odourless gas and burns with a pale blue flame.
- The average composition of CNG is: CH_4 = 70-90%, C_2H_6 = 4-9% & traces of propane and butane.
- Its calorific value is 12500 Kcal /m3.

Uses: It is used:

- as a fuel in low emissive vehicles like ULEV (ultra-low emission vehicles).
- as a domestic and industrial fuel.
- as a source of carbon in tyre industry.
- for the production of H_2 gas needed in fertilizer industry.

Biogas

Biogas a renewable fuel that's produced when organic matter, such as food or animal waste, is broken down by microorganisms in the absence of oxygen. This process is called anaerobic digestion. For this to take place, the waste material needs to be enclosed in an environment where there is no oxygen.

Biogas can occur naturally or as part of an industrial process to intentionally create it as a fuel

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